Teacher:	Ray Joshua Balingkit	**Subject:**	Design & Technology
Week:	3 (Term 1)	**Year:**	10–11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will explain environmental, social and economic challenges in product design. They will develop their understanding through a teacher explanation, a worked example and an evidence-based classroom task. In the second lesson, students will apply a lifecycle perspective to identify and justify a product improvement. Their recorded reasoning, practical evidence and final checks will show whether they can apply the ideas to a defined user, product or design problem.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher questioning, labelled diagrams, case comparisons and the completed independent tasks. Check accurate terminology, evidence linked to the user or brief and a reasoned explanation of choices. Review recorded tests or traces alongside conclusions, and use the final check in both lessons to identify the next skill needing practice.

LESSON SLOTS - TUESDAY AND WEDNESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	8 September 2026	By the end of this lesson, students will be able to explain environmental, social and economic challenges in product design.	**H:** Display a low-cost disposable item and a repairable alternative. Ask what the purchase price leaves out when judging their overall impact. **ID:** Introduce environmental effects such as resource extraction and waste, social issues such as working conditions and accessibility, and economic issues such as production cost and affordability. Explain that these considerations can conflict and that material labels alone do not establish a product's overall impact. **WD:** Analyse a fictional desk lamp with a replaceable light source and a sealed alternative. Model how repair access may extend use but introduce extra parts or assembly costs. Separate the evidence given in the	**Support:** Provide category examples and a benefit-drawback organiser. Use a familiar product and highlight relevant case evidence so students can explain a concrete issue before using wider terminology. **Extension:** Students identify a hidden consequence of an improvement and propose evidence that would help compare the original product with the revised version.

			case from assumptions that would require supplier information. **YD:** Groups sort product-case statements into environmental, social and economic considerations, allowing justified overlap. Individually students propose two improvements to a classroom product and explain one likely benefit and one possible drawback for each. **C:** Ask students to explain one conflict between affordability and durability without assuming all cheaper products are poor quality. Check the proposed improvements for a clear connection to the problem identified.	
2	9 September 2026	By the end of this lesson, students will be able to apply a lifecycle perspective to identify and justify a product improvement.	**H:** Ask where a reusable bottle's story begins and ends. Challenge an answer that starts at purchase or stops when the owner no longer wants it. **ID:** Explain the stages of raw material extraction, processing, manufacture, transport, use and end of life. Introduce reducing, reusing, repairing and recycling as possible interventions. Clarify that a classroom lifecycle map identifies likely impacts and is not a quantified lifecycle assessment. **WD:** Draw a lifecycle map for a bottle using a fictional supply chain. Mark energy use during manufacture, transport distance, washing during use and material separation at disposal. Model how longer use can change the comparison and why recycling depends on local collection and separation.	**Support:** Provide ordered lifecycle cards, a template and one completed stage. Let students annotate a diagram before writing a linked explanation of their chosen intervention. **Extension:** Students compare two improvement options across the lifecycle and explain how product lifespan or local disposal arrangements might alter their recommendation.

			YD: Students construct a lifecycle map for a selected stationery product using supplied information. They mark likely impacts at each stage and select one feasible improvement. They explain its expected effect, any trade-off and the information still needed to judge the overall result. **C:** Ask students to trace how one decision affects more than one stage. Check that they do not claim a precise carbon saving without data or treat recyclable as a guarantee that recycling happens.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel GCSE (9–1) Design and Technology revision notes: 1.14 Environmental, social and economic challenges; 1.1 Sustainability
- Teacher computer, display screen and whiteboard markers
- Prepared product scenarios, diagrams and user briefs aligned to the lesson objectives
- Prepared analysis worksheet, comparison grid or test-recording table
- Paper, pencils, rulers, card templates and approved classroom stationery
- Teacher-prepared samples or component cards; pre-cut model parts where required
- Lesson success criteria, observation checklist and labelled evidence folders



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?